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Emotional effects of startling background music during reading news reports: The moderating influence of dispositional BIS and BAS sensitivities

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We examined the moderating influence of dispositional behavioral inhibition system (BIS) and behavioral activation system (BAS) sensitivities on the relationship of startling background music with emotion-related subjective and physiological responses elicited during reading news reports, and with memory performance among 26 adult men and women. Physiological parameters measured were respiratory sinus arrhythmia (RSA), electrodermal activity (EDA), and facial electromyography (EMG). The results showed that, among high BAS individuals, news stories with startling background music were rated as more interesting and elicited higher zygomatic EMG activity and RSA than news stories with non-startling music. Among low BAS individuals, news stories with startling background music were rated as less pleasant and more arousing and prompted higher EDA. No BIS-related effects or effects on memory were found. Startling background music may have adverse (e.g., negative arousal) or beneficial effects (e.g., a positive emotional state and stronger positive engagement) depending on dispositional BAS sensitivity of an individual. Actual or potential applications of this research include the personalization of media presentations when using modern media and communications technologies.

Key words: Music, startle response, psychophysiology, emotion, personality traits.

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It is well recognized that music is a powerful elicitor of emotion (e.g., Goldstein, 1980; Panksepp, 1995), and musical compositions have been shown to prompt similar ratings of basic emotions (i.e., happiness, sadness, fear, and anger) across different listeners (e.g., Cupchick, Rickert & Mendelson, 1982; Krumhansl, 1997). Different types of music have different effects on arousal levels and anxiety, and music can be used to promote relaxation (e.g., Davis & Thaut, 1989; Iwanaga, Ikeda & Iwaki, 1996; Stoudenmire, 1975).

A variety of studies have demonstrated that different types of music induce varying emotion-related psychophysiological responses (e.g., Iwanaga & Moroki, 1999; Savan, 1999). Krumhansl (1997) found that sad excerpts produced the largest changes in heart rate (HR), blood pressure, electrodermal activity (EDA), and digital temperature; fear-related excerpts produced the largest changes in pulse transit time and finger pulse amplitude; happy excerpts produced the largest changes in respiration measures. Thayer and Levenson (1983) found, in turn, that music added to stressful films elicited changes in EDA. Gerra *et al*. (1998) showed that HR and systolic blood pressure increased during techno-music, but not during classical music. They also found that emotional and neuroendocrine responses during techno-music were associated with Cloninger's temperament dimensions (i.e., Novelty Seeking and Harm Avoidance); that is, these temperament dimensions moderated the relation of music to emotional responses. It has also been shown that rock music induced a decrease in respiratory sinus arrhythmia (RSA; i.e., high-frequency component of HR variability) indexing cardiac parasympathetic activity (McCraty, Atkinson, Rein & Watkins, 1996). There is also evidence for a dissociation between subjective and physiological emotional responses to music. Davis and Thaut (1989) showed that, although music decreased state anxiety, it aroused and excited rather than soothed autonomic nervous system and muscular activity.

The influence of different characteristics of music on emotions and psychophysiological responses has not been investigated thoroughly, however. Abrupt changes and contrasts in dynamics (i.e., potentially startling stimulus features) may be one interesting musical characteristic in this connection. Seventeenth-century theorists linked dynamics to the rhetorical musical figures associated with the doctrine of the Affections (Sadie, 2001). For example, according to Michael Praetorius, '*pian* and *forte*' are methods used 'to express the *affectus* and move human feelings'. Powerful dynamic contrasts are also present in the works of composers of the Classical era (e.g., C. P. E. Bach, F. J. Haydn, W. A. Mozart, and L. van Beethoven), and postmodern composers (e.g., Ligeti) tended to employ extremes at both ends of the dynamic spectrum (Sadie, 2001). At present, startling dynamic contrasts are often present in thriller music. A good example of a musical composition with startling stimulus features is Haydn's Symphony No. 94 in G major ('Surprise'), H. 1/94 2nd movement (Andante). In the Andante, there are sudden kettledrum beats (fortissimo), and it has been

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documented that Haydn had said: 'there the ladies will jump' (Robbins, 1988).

It may be of interest to examine the emotional impact of music with such sudden fortissimo chords, given that sudden fortissimo chords may elicit so-called startle or defense responses in the listener. The startle response has been suggested as being elicited by intense (> 80 dB), transient, and rapid rise-time stimuli, although there is also evidence that it may be elicited by a broader range of stimuli (Turpin, Schaefer & Boucsein, 1999). The startle response is represented by cardiac acceleration, changes in RSA, greater amplitudes and frequencies of electrodermal responses, digital vasoconstriction, and eyeblinks (Reyes del Paso, Godoy & Vila, 1993; Turpin, 1986; Turpin *et al.*, 1999). Although the startle response has been extensively studied in psychophysiological basic research, it has been suggested that the context (e.g., music) in which the startle response is elicited may alter the psychophysiological response (Cook & Turpin, 1997).

Music has also been shown to influence the performance of cognitive tasks. Rauscher, Shaw, and Ky (1993, 1995) found significant enhancement on spatial-temporal tasks after listening to a Mozart piano sonata (i.e., Mozart effect; see also Chabris *et al.*, 1999). Likewise, in the study by Cockerton, Moore, and Norman (1997), background music facilitated cognitive test performance. In contrast, Ransdell and Gilroy (2001) reported that background music significantly interfered with word-processed writing; they suggested that unattended music may place heavy demands on working memory. It has also been shown that increasing complexity in background music is associated with an increase of extroverts', and a decrease of introverts', cognitive-task (e.g., memory test) performance (Furnham & Allass, 1999). Although one might speculate that startling background music is likely to interfere with cognitive performance, it could also have beneficial effects. For example, startling background music would be expected to elicit arousal, and increases in arousal have been suggested as resulting in additional capacity being allocated to the task of storing messages in long-term memory (e.g., Lang, Newhagen & Reeves, 1996; Revelle & Loftus, 1992; see also Vila, Pérez, Fernández, Pegalajar & Sánchez, 1997).

As noted above, there may be important individual difference variables, i.e., moderator variables, that might affect the nature and magnitude of relations of music with emotional responses and cognitive performance. The behavioral inhibition system (BIS) and behavioral activation system (BAS) are the primary brain motivational systems underlying behavior and affect (Gray, 1982, 1991; Pickering & Gray, 1999). The BIS and BAS have also been suggested to underlie Harm Avoidance and Novelty Seeking temperament traits, respectively, that have previously been found to moderate the influences of music (Cloninger, 1987; Gerra *et al.*, 1998). Likewise, introversion-extraversion is a derived dimension reflecting the balance between the reactivities of the BIS and BAS (Gray, 1991). Therefore, dispositional BIS and BAS sensitivities are logical candidates for moderator variables when examining the influence of startling music on emotions and performance (see also Zuckerman, 1990). The BIS is activated by conditioned stimuli signaling punishment or frustrative nonreward, and by innately fear-provoking or novel stimuli (Gray, 1991; Pickering & Gray, 1999); it is associated with increased (negative) arousal and attention. The BAS is activated by conditioned stimuli signaling reward or relief from punishment. Greater BAS sensitivity is reflected in greater responsiveness to, and engagement with, appetitive stimuli (Gray, 1991; Pickering & Gray, 1999). Both the BIS and BAS are presumed to be related to a broad affective quality, i.e., negative affect and positive affect, respectively (e.g., Gray, 1991). It is also of note that EDA and HR are strongly associated with BIS and BAS activity, respectively (Fowles, 1980).

In view of the above considerations, the aim of the present investigation was to examine the influence of startling background music on emotion-related subjective and physiological responses elicited during reading news reports, and on memory performance. It was also asked whether dispositional BIS and BAS sensitivities moderate the potential influences of music. It would be expected, for example, that individuals scoring high on the BIS dimensions would react more strongly (e.g., greater negative arousal) to startling background music (i.e., a signal of punishment and novelty). In the current study, several psychophysiological measures were included: (a) EDA indexing emotional and sympathetic arousal, (b) facial electromyography (EMG) indexing positive and negative emotions, and (c) RSA indexing attention and emotions.

METHODS

Subjects

Subjects were 26 Finnish non-musician men (n = 10) and women with varying educational backgrounds, who ranged from 20 to 62 years of age (M = 32). They participated in return for two movie tickets. Given that physiological data were unusable for some subjects for varying reasons, the number of subjects ranged from 21 to 26 in the statistical analyses.

Stimulus materials

Background music. As background music, we used Franz Joseph Haydn's Symphony No. 94 in G major ('Surprise'/'the Drumstroke'/ 'Mit dem Paukenschlag'), H. 1/94 2nd movement (Andante). In this excerpt, there are intermittent, abrupt kettledrum fortissimos. We made two versions from this excerpt. In one version, these abrupt kettledrum fortissimos were modified in some degree using Sound Forge audio editing software: (a) they were intensified so that the peak sound intensity was about 88–90 dB SPL (Udeyraj & Sons sound meter, Model AZ 8928) at the point of the subject's ears (about 115 cm above the floor and 120 cm from the loudspeakers) and (b) the rise time was shortened to be below 30 ms. These modifications were done to ensure that the kettledrum fortissimos

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would elicit a startle response; despite the modifications, the excerpt sounded natural. The kettledrum fortissimos (peak sound intensities) occurred at 1.990 s and 18.135 s from the beginning of the excerpt, and the duration of each fortissimo was slightly over one second. In another version, these kettledrum fortissimos were totally removed. With the exception of the kettledrum fortissimos, the two versions were identical and the mean sound intensity was about 60 dB (about 60 bpm) in both of them. The duration of the excerpts was about 66 s (bars 32–65), and they were presented through Harman/Kardon multimedia loudspeakers.

News messages. The reading task stimuli consisted of six financial news stories that were selected on the basis of a pre-test procedure. For the pre-test procedure, 44 financial news stories in written form were selected from Kauppalehti Online in the Internet (http://www.kauppalehti.fi). News stories that referred to a very significant and time-bound event were excluded. In addition, the selected stories were edited to conceal the fact that they were not news of the day. Seventeen pre-test subjects who did not participate in the study rated the 44 stories on affective dimensions of valence and arousal using nine-point pictorial scales presented on a computer screen (see below). On the basis of median splits of these ratings, six stories representing the same valence-arousal combination (i.e., negative-high arousal) were selected for the present study (e.g., Jealousy distorts wage policy, Where goes taxation in the new millennium?). This was done to eliminate any potential confounding influence of emotional content, although the (emotional) content of the stories was not considered to be of prime importance in the present study. As a general rule, financial news stories are not likely to cover the full range of affective reactions. Each of the six news stories had two versions, one accompanied by background music with kettledrum fortissimos (i.e., startling music) and one by background music without kettledrum fortissimos (i.e., non-startling music). The news stories were of approximately equal length (81–87 words).

The subjects read the six news stories in one of two presentation orders. In each order, three of the stories were accompanied by startling background music and three by non-startling music. To control order of presentation, each story appeared in a different position (in the two orders) and was preceded and followed by different stories. Each story appeared as a startling version in one order and as a non-startling version in the other order. One order began with a news story with startling music while the other order began with a news story with non-startling music, and the stories with startling music alternated with stories with non-startling music.

This study belongs to a series of studies examining news reading using pocket PCs. Therefore, the news stories were presented on the screen of a simulated Casio Cassiopeia E-115 Pocket PC (the technical properties of the present pocket PCs made their use impracticable). The pocket PC was simulated on a 10.4″ TFT XGA display of a Fujitsu Stylistic 3400X Pen Tablet. The simulated pocket PC was of natural size (screen size was 60 mm [width] × 80 mm [height]) and appeared in the middle of the Pen Tablet display, the rest of the display being black.

Measures

Valence and arousal. Subjects rated their emotional reactions in terms of valence and arousal to each of the news stories using nine-point pictorial scales presented on a computer screen. The valence scale consists of nine graphic depictions of human faces in expressions ranging from a severe frown (most negative) to a broad smile (most positive). Similarly, for arousal ratings, there are nine graphical characters varying from a state of low visceral agitation to that of high visceral agitation. The ratings are made by selecting a radio button below an appropriate picture. These scales resemble Lang's (1980) Self-Assessment Manikin.

Interest ratings. The subject rated interest in the news message using a five-point scale, ranging from 1 (not interesting at all) to 5 (very interesting). The scale was presented on a computer screen.

Knowledge acquisition. A set of 18 multiple-choice questions (three questions per news story) was used to assess the amount of knowledge acquired from the stories. The items focused on important facts central to the story. An effort was made to make the multiple-choice questions completely text dependent. There was one correct answer to each question, and three plausible alternatives were constructed for each question. The questions were rearranged for each reading order so that items about each story were presented in the order that the stories were read.

BIS and BAS sensitivities. Dispositional BIS and BAS sensitivities of the participants were measured with the BIS/BAS scales (Carver & White, 1994), a 20-item self-administered questionnaire. The BIS scale is comprised of seven items (e.g., 'I feel pretty worried or upset when I think or know somebody is angry at me'). The BAS scale is comprised of three subscales: Drive (4 items; e.g., 'I go out of my way to get things I want'), Reward Responsiveness (5 items; e.g., 'When I get something I want, I feel excited and energized'), and Fun Seeking (4 items; e.g., 'I crave excitement and new sensations). Each of the items was rated on a four-point scale, ranging from 1 (very false with me) to 4 (very true for me). The psychometric properties of the instrument have been shown to be acceptable (Carver & White, 1994).

Physiological data collection

Electrocardiogram (ECG) was recorded using the Psylab Model BIO2 isolated AC amplifier (Contact Precision Instruments, London, UK). Three Red Dot Ag/AgCl electrodes (3M Health Care, Borken, Germany) were placed in a modified Lead II position.

Skin conductance was recorded with the Psylab Model SC5 24 bit digital skin conductance amplifier that applied a constant 0.5 V across Ag/AgCl electrodes with a contact area of 8 mm diameter (Med Assoc. Inc., St. Albans, VT). Electrodes were filled with TD-246 skin conductance electrode paste (Med Assoc. Inc.) and attached to the middle phalanges of the first and second fingers of the subject's nondominant hand after hands were washed with soap and water.

Facial EMG activity was recorded from the left corrugator supercilii and zygomatic major muscle regions as recommended by Fridlund and Cacioppo (1986), using surface Ag/AgCl electrodes with a contact area of 4 mm diameter (Med Assoc. Inc.). Electrodes were filled with TD-240 electrode gel (Med Assoc. Inc). The raw EMG signal was amplified, and frequencies below 30 Hz and above 10 kHz were filtered out, using the Psylab Model EEG8 amplifier. The raw signal was rectified and integrated using the Psylab INT8 contour following integrator (time constant = 50 ms).

The digital data collection was controlled by Psylab7 software, and all physiological signals were sampled at a rate of 500 Hz.

Procedure

After a brief description of the experiment, the subject filled out an informed consent form. Electrodes were then attached and the subject was seated in a comfortable armchair. During a rest period of 7 min, baseline physiological measurements were performed. Fujitsu Stylistic Pen Tablet was placed on a stand on a table in front of the subject. The subject was instructed that he or she should initiate the presentation of each news story by clicking an appropriate link in a list using a mouse (there was a separate list of links for

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each of the two presentation orders). The subject scrolled the text with the mouse. After each story, the subject rated the story on several dimensions (e.g., valence, arousal, interest). After reading all stories, the subject was given a questionnaire measuring the amount of knowledge acquired from the stories (the subject was informed about this before reading the stories) and a number of other questionnaires. The room was dimly illuminated during the baseline and when reading the news stories.

Data analysis

To quantify RSA, the ECG data were analyzed off line by using WinCPRS software (Absolute Aliens Ay, Turku, Finland). The R-peaks were detected using a digital filter and fitted with a polynomial of a second order. After deriving an R-R interval series, the R-peak detection results on the ECG signal were edited manually and the R-R interval series was automatically updated. The interbeat interval (IBI) data were then resampled into equal time intervals every 200 ms. The non-stationary component was removed from the IBI time series by using a cubic order moving polynomial filter with a duration of 10.5 s, and a residual series was generated (Porges and Bohrer, 1991). All later operations and computations were performed separately for the first 30-second residual data segment of each news story. The data segments were windowed (Hanning window) and subjected to spectral analysis using the fast Fourier transform (FFT) technique. The computed spectrum was smoothed using a triangular averaging function. RSA was indexed by the spectral power for the 0.15–0.40 Hz frequency band. The RSA data were transformed with a natural logarithm and reported in units of $\ln(\text{ms})^2$.

Mean values for facial EMG activity were derived for each of the first six 5-second epochs during the news stories, and for the last three minutes during the baseline. In addition, skin conductance fluctuations exceeding 0.02 μS were scored as non-specific skin conductance responses (NS-SCRs). The frequency of NS-SCRs was calculated for the first six 5-second epochs during the news stories. A log transformation (log [NS-SCR + 1]) was performed to normalize the distribution.

All data were then analyzed by the General Linear Model (GLM) Repeated Measures procedure in SPSS, with continuous independent variables (i.e., BIS and BAS), each in turn, as a covariate. The analyses included two or three within-subject factors, i.e., type of background music (2 levels: startling, non-startling), story position (3 levels: 1st, 2nd, 3rd), and time (for EDA and EMG, 6 levels: the first six 5-second epochs during news reading). In addition, age and the baseline physiological value were used as covariates when analyzing RSA (age may influence RSA), and the baseline physiological value was used as a covariate when analyzing facial EMG. The multivariate analyses were used for the repeated measures data. Linear trend components within these analyses were used to delineate effects across levels. The three levels of the story position factor refer not to specific story content but to the position of a story with a given type of background music (startling or non-startling) relative to the other stories with the same type of background music. This is done to maximize effects due to type of background music and time and minimize topic effects, given that unpredictable topic effects were of no particular interest here. The impact of startling background music may develop across repeated exposures to stories with startling music (and/or across time within a single story); repeated exposures to stories with startling music could lead to sensitization or desensitization (or habituation) of responses, or they also may simply be maintained across the exposures (see Geyer & Braff, 1987). We carried out the analyses also with gender as a between-subjects factor; however, given that gender turned out to play no significant role in the present connection, it was omitted from the final analyses. Significant interactions were graphed for participants receiving low (1 *SD* below the mean) and high (1 *SD* above the mean) scores on the moderator variable (e.g., BAS score; see Aiken & West, 1992).

RESULTS

Affective dimensions and interest ratings

The GLM Repeated Measures analysis revealed a significant main effect for type of background music in predicting pleasure ratings for the news stories, $F(1, 22) = 4.46$, $p = 0.046$, $\eta^2 = 0.17$. Pleasure ratings were lower for news stories with startling music as compared to those with non-startling music, *M*s = 4.67 and 5.33, respectively. The interaction between BAS Reward Responsiveness and type of background music failed to reach statistical significance in predicting ratings of pleasure, $F(1, 22) = 2.98$, $p = 0.098$, $\eta^2 = 0.12$. However, as illustrated in Figure 1 (top panel), news stories accompanied by music with startling characteristics tended to be rated as less pleasant than those accompanied by non-startling music among low Reward Responsiveness scorers, but not among high Reward Responsiveness scorers. There was also a significant main effect for type of background music when predicting arousal ratings for the news stories, $F(1, 22) = 6.08$, $p = 0.022$, $\eta^2 = 0.22$. Arousal ratings were higher for news stories with startling music as compared to those with non-startling music, *M*s = 4.89 and 4.78, respectively. In addition, there was a significant BAS Reward Responsiveness × Type of Background Music interaction in predicting arousal ratings of news stories, $F(1, 22) = 5.81$, $p = 0.025$, $\eta^2 = 0.21$. Figure 1 (middle panel) shows that news stories accompanied by startling music were rated as more arousing than news stories accompanied by non-startling music among low Reward Responsiveness scorers, but not among high Reward Responsiveness scorers.

The GLM Repeated Measures analysis revealed a significant main effect for type of background music in predicting ratings of interest in news stories, $F(1, 22) = 4.76$, $p = 0.040$, $\eta^2 = 0.18$. News stories accompanied by startling music were rated as more interesting as compared to news stories with non-startling music, *M*s = 3.25 and 3.13, respectively. In addition, there was a significant interaction between BAS Drive and type of background music in predicting interest ratings, $F(1, 22) = 6.17$, $p = 0.021$, $\eta^2 = 0.22$. Figure 1 (bottom panel) illustrates that news stories accompanied by startling music were rated as more interesting than news stories accompanied by non-startling music among high Drive scorers, but not among low Drive scorers. There were no significant main effects for, or interactions involving, the story position factor in predicting the message ratings.

Electrodermal activity

The GLM Repeated Measures procedure revealed a main effect for type of background music in predicting NS-SCR

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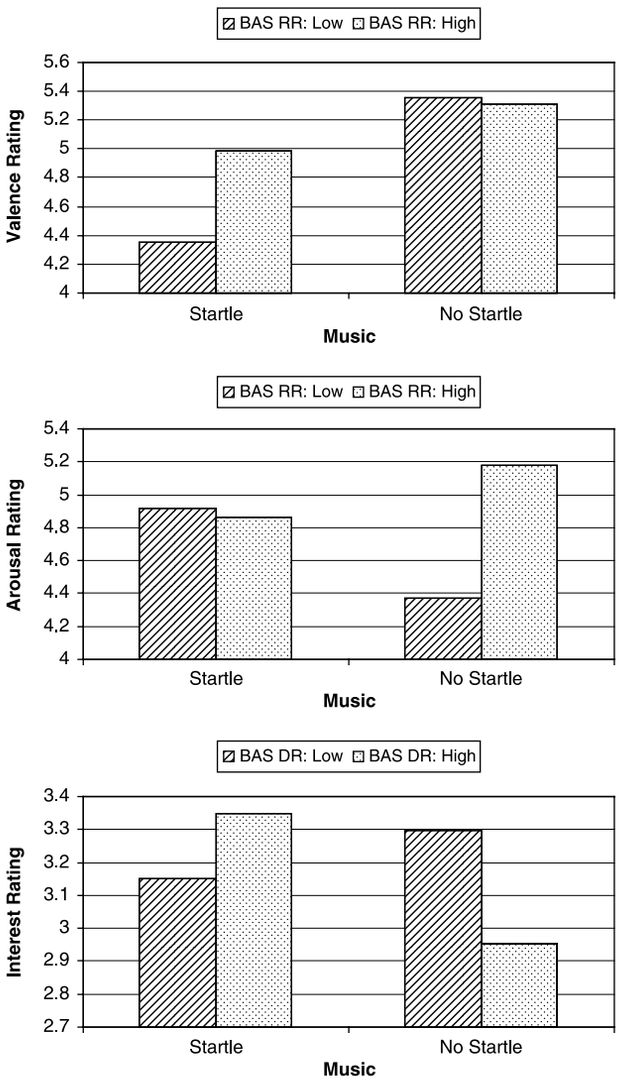

Fig. 1. Valence (top panel) and arousal (middle panel) ratings for news stories with startling and non-startling background music as a function of BAS Reward Responsiveness (BAS RR). Interest ratings for news stories with startling and non-startling background music as a function of BAS Drive (BAS DR; bottom panel).

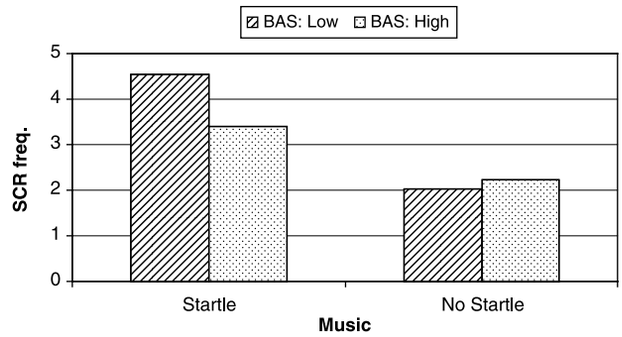

Fig. 2. The frequency of non-specific skin conductance responses for news stories with startling and non-startling background music as a function of BAS sensitivity.

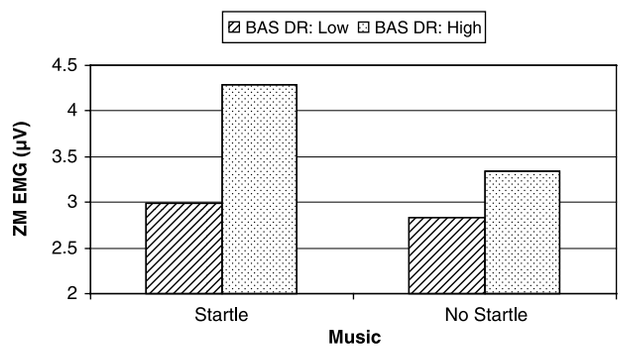

Fig. 3. Zygomatic EMG activity for news stories with startling and non-startling background music as a function of BAS Drive (BAS DR).

freq. during news reading, $F(1, 24) = 10.46$, $p = 0.004$, $\eta^2 = 0.30$. NS-SCR freq. was higher when reading news stories with startling music as compared to non-startling music, *M*s = 3.97 and 2.12, respectively. The Multivariate analysis revealed also significant Global BAS × Type of Background Music and BAS Fun Seeking × Type of Background Music interactions for NS-SCR freq. during news reading, *F*s(1, 24) = 5.64 and 12.71, *p*s = 0.026 and 0.002, η^2s = 0.19 and 0.35, respectively. Figure 2 shows that startling background music prompted a more pronounced increase in NS-SCR freq. among low Global BAS scorers as compared to high Global BAS scorers. This was the case also for BAS Fun Seeking. The main effects for, and interactions involving, the within-subjects factors of story position and time were non-significant.

Facial EMG

The GLM Repeated Measures analysis showed a significant BAS Drive × Type of Background Music interaction in predicting zygomatic EMG activity, $F(1, 18) = 4.95$, $p = 0.039$, $\eta^2 = 0.22$. Figure 3 illustrates that individuals scoring high on BAS Drive exhibited pronouncedly high zygomatic EMG activity when reading news stories with startling background music. Other main effects and interactions were non-significant.

RSA

The GLM Repeated Measures analysis revealed a significant BAS Reward Responsiveness × Type of Background Music interaction in predicting RSA, $F(1, 20) = 4.98$, $p = 0.037$, $\eta^2 = 0.20$. Figure 4 illustrates that startling background music induced an increase in RSA among high BAS Reward Responsiveness scorers. Other main effects and interactions were non-significant.

Reading time

There was a significant main effect for type of background music in predicting reading time, $F(1, 25) = 10.95$, $p = 0.003$,

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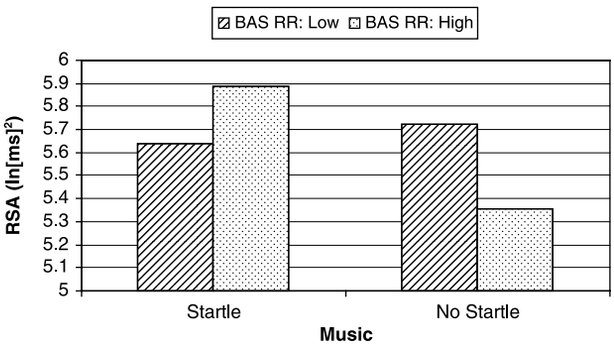

Fig. 4. Respiratory sinus arrhythmia (RSA) for news stories with startling and non-startling background music as a function of BAS Reward Responsiveness (BAS RR).

$\eta^2 = 0.31$. Reading times were longer during startling background music as compared to non-startling background music, *M*s = 40.8 s and 36.6 s, respectively.

Memory for news stories

There was no significant main effect for type of background music on the number of correct answers to multiple choice questions, $F(1, 25) = 0.82$, $p = 0.373$, $\eta^2 = 0.03$. This was not changed when an adjustment was made for reading time, $F(1, 24) = 0.70$, $p = 0.410$, $\eta^2 = 0.03$. The interactions of BIS and BAS scores with the type of background music were also non-significant. The number of correct answers correlated positively and significantly with reading time, $r = 0.41$, $p = 0.040$.

DISCUSSION

In the present investigation, we examined the influence of startling background music on emotion-related subjective and physiological responses elicited during reading news reports, and on memory performance. We also examined whether dispositional BIS and BAS sensitivities moderate these influences. The results revealed several main effects of startling music: news stories with startling music were rated as less pleasant, more arousing, and more interesting, and they also prompted higher physiological arousal as measured by NS-SCR freq. These main effects were qualified by the significant interaction of BAS × Type of Music, however. Dispositional BIS sensitivity appeared to play no significant role here; this is unexpected, given that the BIS has been suggested as being sensitive to novelty and high-intensity stimuli (Corr, Pickering & Gray, 1995). No effects indicative of response habituation or sensitization were found either.

Our main finding was that, among high BAS (Drive) scorers, news stories with startling background music were rated as more interesting and prompted higher zygomatic EMG activity than news stories with non-startling music. Thus, among individuals with high BAS sensitivity, relatively uninvolving news stories are experienced as more interesting and a more positive emotional state ensues as a result of startling background music, given that zygomatic EMG activity indexes positive emotions (e.g., Lang, Greenwald, Bradley & Hamm, 1993). This suggestion is also partly supported by the finding that news stories with startling music were rated as less pleasant among low BAS (Reward Responsiveness) scorers, but not among high BAS (Reward Responsiveness) scorers. These findings suggest that high BAS individuals show 'approach'-oriented, as opposed to 'withdrawal'-oriented, emotional responses during startling background music, given that high BAS sensitivity increases the probability of approach behavior (Gray, 1991). Recall also that extraverts and high sensation seekers show a tendency to stimulation seeking in order to raise their arousal to the hedonic optimal (see Matthews & Gilliland, 1999; Zuckerman, 1990); Gray's BAS (or impulsivity) dimension is closely aligned with extraversion (rotation of approx. 30°), and high BAS activity underlies the expression of the sensation-seeking trait. High sensation seekers have also been shown to have high thresholds for intense auditory stimulation, and to like rock music, for example (Zuckerman, 1990). We have also previously found in another data set that BAS sensitivity is positively associated with acoustic startle stimuli-induced changes in Activated Pleasant Affect (unpublished data, 2002; see also Heponiemi, Keltikangas-Järvinen, Puttonen & Ravaja, 2003).

It was also found that, among low BAS scorers, startling background music elicited an increase in arousal ratings for the news stories and in NS-SCR freq. indexing sympathetic arousal. In addition, they rated the news stories with startling background music as less pleasant. Together these findings suggest that individuals with low BAS sensitivity were negatively aroused by startling background music. It is puzzling, however, why high BAS (Reward Responsiveness) scorers rated the news stories with non-startling music as highly arousing. This is probably a chance finding, although low arousal-inducing environments (e.g., when the subject is simply lying with eyes closed) have been suggested to result in paradoxical arousal in extraverts (cf. high BAS scorers; Matthews & Gilliland, 1999).

We also found that startling background music elicited an increase in RSA among individuals with high BAS (Reward Responsiveness) scores. Gianaros and Quigley (2001) have previously found that moderate intensity pure tone stimuli (75 dB SPL, 1.5 s duration) elicited an increase in RSA; the authors suggested that the stimulus characteristics likely fell on the border between a typical orienting response and a typical defense response. Likewise, in another data set, we found that an auditory startle stimulus (distorted sound of 400 Hz, 109 dB, instantaneous risetime, and 0.5 s duration) prompted an increase in RSA (1-min mean; unpublished data, 2002). In contrast, as noted above, rock music (about 70 dB) has been found to induce a decrease in RSA (McCraty, Atkinson, Tiller, Rein & Watkins, 1996). When

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examining the RSA response pattern associated with the cardiac defense response to intense auditory stimulation, Reyes del Paso *et al.* (1993) suggested that the cardiac defense response is composed of four components, two reductions and two increases, that run parallel, but in opposite direction, to the HR changes.

It is of note that, when there are no or minimal explicit task demands, attentional stimuli (e.g., visual illusion task) yield an independent activation of vagal control of the heart as indexed by increased RSA (Berntson, Cacioppo & Fieldstone, 1996), although the direction of change in RSA during stimulus orienting and sustained attention is a controversial issue (for a potential explanation, see Richards & Casey, 1991). Thus, increased RSA during startling background music among high BAS (Reward Responsiveness) scorers might indicate that startling background music increases their attention. However, it should be recognized that the type of background music was unrelated to memory performance. It has also previously been suggested that negative and positive emotions are related to decreased and increased RSA, respectively (McCraty *et al.*, 1996; Porges, 1995). Thus, to the extent that this suggestion holds, the RSA data might suggest, as did the EMG data, that startling background music elicits a more positive emotional state among individuals with high BAS sensitivity. It might also be suggested that the emotional influences of music with abrupt changes and contrasts in dynamics are based on a violation of the law of good continuation (i.e., one of the so-called gestalt laws). That is, even short and simple musical sequences engender powerful expectancies in listeners for what will follow these sequences (e.g., Carlsen, 1981; Krumhansl, 1996). Importantly, it has been suggested that the confirmations and violations of these expectancies that may be based on the fundamental properties of human perception, such as the gestalt laws, are responsible for some emotional responses to music (Krumhansl, 1996; Narmour, 1990).

In sum, the present study showed that the relationship of startling background music with emotion- and attention-related subjective and physiological responses elicited during reading news reports differed as a function of dispositional BAS sensitivity of an individual. Startling background music elicited negative arousal among low BAS individuals, whereas a positive emotional state and stronger positive engagement ensued among high BAS individuals, although background music had no effect on memory performance. Thus, the present study suggests that a given type of background music may have adverse or beneficial effects depending on dispositional BAS sensitivity of an individual. Given that we manipulated only one characteristic of music, future studies might manipulate other musical features. Future studies should also compare the musical background with a non-musical one, or compare different types of music involving startling stimulus features. Actual or potential applications of the present research include the personalization of media presentations when using modern media and communications technologies, among other things (see Riecken, 2000). That is, the background music of a media presentation can be adapted to fit the individual characteristics (e.g., BAS sensitivity) of the user, thereby eliciting more positive engagement.

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